



PATTERN AI LABS

EDITION • 2026 • RETAIL & QSR

— EDGEVISION • STORE FLOORS & QUICK-SERVICE

EdgeVision

Retail & QSR

Every camera, a real-time operations engine.

Intelligent AI surveillance at the edge — turning every store and restaurant camera into a real-time operations & customer-intelligence engine. One on-prem edge platform across retail floors and quick-service restaurants.

● EDGE-FIRST ● PRIVACY BY DESIGN ● INTEL POWERED

EDGEVISION.PRO • PATTERN AI LABS

RETAIL
& QUICK-SERVICE

CHAPTER 01

The Visibility Gap

Modern retail and QSR sites capture thousands of hours of CCTV every week.
Almost none of it becomes a decision.

— THE CHALLENGE · STORE FLOOR

The In-Store Visibility Gap.

Stores capture hours of CCTV every shift. But almost none of it becomes a decision.

- Staff effort & productivity is opaque — who is attending, grouping, on the phone
- Customer journey is invisible after entry — footfall $\neq$ engagement
- High-value customers walk in unannounced — loyalty runs blind to the first 60 seconds
- Floor-hygiene failures surface late — scattered footwear, disturbed displays, > 4 in back store



CHAPTER 02

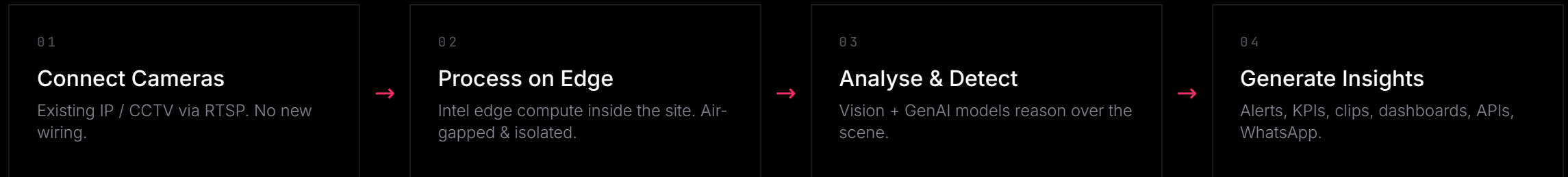
The Platform

A single edge appliance per site. Hybrid Vision + GenAI on-prem. Insights in seconds.

— THE SOLUTION

EdgeVision Platform.

High-performance Intel edge compute on the site LAN ingests existing CCTV, runs hybrid Vision + GenAI on-prem, and produces real-time alerts, dashboards and APIs.



Real-time

DETECTION

Multi-cam

PER EDGE NODE

0_{KB}

CLOUD BANDWIDTH

<1_{wk}

DEPLOYMENT

LIVE OPERATIONS CONSOLE

One Pane of Glass for the Store Floor.

SNAPSHOT

FOOTFALL

1,284

ACTIVE STAFF

12 / 14

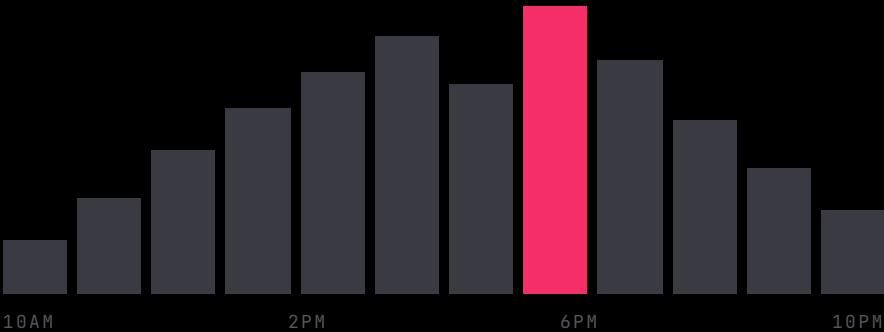
ALERTS

3

AVG DWELL

4:21

HOURLY FOOTFALL



Hour	Footfall
10AM	100
11AM	150
12PM	200
1PM	250
2PM	300
3PM	350
4PM	300
5PM	400
6PM	450
7PM	350
8PM	250
9PM	150
10PM	100

ALERT STREAM

- VIP entered · gold tier
- Wall #3 unattended 2m
- Queue > 4 at till 2
- Back store > 4 staff

CAM 01

CAM 02

CAM 03

CAM 04

CAM 05

CAM 06

CHAPTER 03

Two Lenses, One Network

The same feeds answer two questions at once: what is the customer experiencing, and how is the team performing?

— TWO LENSES

Customer & Staff Analytics.

CUSTOMER ANALYTICS

What customers do, where they linger, who they are.

- Heat-mapping by entry, fixture and category
- Walk-in flow between departments and categories
- Self-service vs assisted journey detection
- VIP / loyalty member recognition at entry
- POS ↔ camera time-stitch for billing intelligence

STAFF ANALYTICS

Who is attending, who is missing, who is grouping.

- ✓ Peak vs non-peak presence by zone
- ✓ Grouping > 1 min, mobile use, back-store time
- ✓ Style/size run timing — wall to back-store
- ✓ Per-staff customer attention & conversion proxy
- ✓ Brand-level heat-map for back-store activity

Same hardware. Same cameras. Two outcomes. Intel edge compute on the store LAN — no extra installation.

IN-STORE DETECTION

Tracking Customers & Staff Simultaneously.

Every person on the floor is detected, classified and tracked — without ever storing a face.



— CAPABILITY 01

Customer Analytics, Elaborated.

1

Heat Mapping

By entry, fixture, category. Walk-in flow between categories becomes a measurable cross-category insight.

2

Customer Tracking

VIP & loyalty recognition at entrance. Auto-enroll dwell > 15 min. No biometric data leaves the store.

3

Self-Service vs Assisted

Identifies customers who buy without staff help — critical for VM, range planning and staffing.

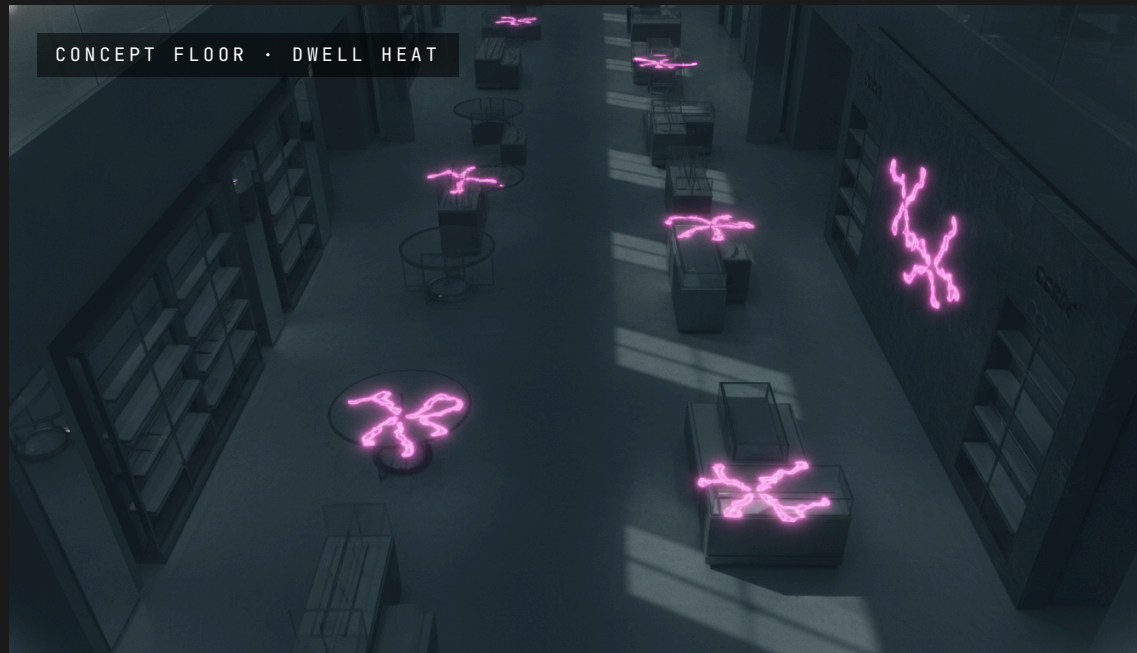
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POS ↔ Camera Stitch

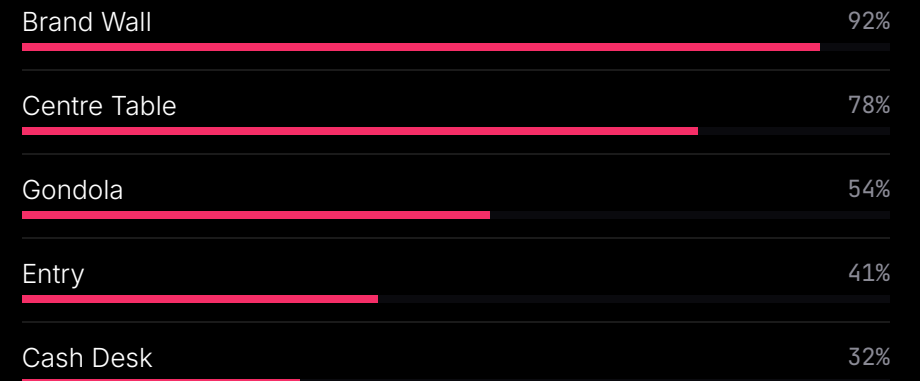
Correlate bills with in-store trail. Every transaction becomes a journey: dwell, attended-by, time-to-bill.

HEAT MAPPING

Customer Dwell, Visualised.



TOP DWELL ZONES



Generated entirely on the in-store edge box. No cloud round-trip.

VIP RECOGNITION FLOW

From CCTV Frame to Manager's Phone.

A loyalty member walks in. Within 3 seconds the manager has identity, history and a recommended action — without a single cloud round-trip.



01

0.4s

Recognition

Privacy-preserving embedding match against the store-local gallery. No raw faces leave the box.



02

2.1s

Notification

Manager pinged via WhatsApp / app — name, tier, last visit, last basket, preferences.



03

3.0s

Activation

Next-best-action: assign senior associate, push targeted POS offer, trigger follow-up sequence.

— CAPABILITY 02

Staff Analytics, Elaborated.

1

Grouping & Inattention

Flags clusters of staff grouped > 1 minute without an active customer. Manager receives a real-time alert with clip.

2

Wall-Location Coverage

Detects when a staffed wall is unattended beyond threshold during peak hours. Trended hourly, store by store.

3

Mobile-Use Tracking

Phone use detected with thresholds (1 min floor, 2 min back store). A coaching tool, not a surveillance tool.

— **CAPABILITY 02 · CONTINUED**

The Hidden Loops That Decide a Sale.

STYLE / SIZE RUN-TIME

- Time between staff leaving the brand location to look for a size and returning with stock
- Measured per-brand, per-staff, per-shift

PER-STAFF SALE CONVERSION

- ✓ Stitching the journey to attending staff and POS bill
- ✓ Yields an objective attended-conversion score

PER-STAFF ACTIVITY · S04 · SHIFT 10-8



— **BRAND-LEVEL STAFF HEAT MAP**

Where Your Staff Actually Spend Time.

FLOOR MOVEMENT MAP

- ✓ Per-staff movement trail across the floor
- ✓ Brand-zone occupancy heat by 30-min slot
- ✓ Hand-off detection between brand zones
- ✓ Comparison: peak vs non-peak coverage

BACK-STORE ACTIVITY

- ✓ Round-trips per brand per day (size hunts)
- ✓ Time spent by each staff in back store
- ✓ Alert if > 4 staff in back store at once
- ✓ End-of-day back-store occupancy report

— **MORE OUT-OF-THE-BOX · RETAIL**

The Long Tail of Use-Cases.

01 Floor & Display Hygiene Footwear out of place > 2 min triggers alert Restore-display task auto-assigned Daily hygiene compliance score per zone End-of-day VM delta photos	02 Queue & Cash-Desk Live queue length & wait time per till Auto-alert when queue > 4 customers Bagging / hand-off duration tracking Trial-room turnover & queue at fitting bay	03 Loss Prevention Tag-removal & concealment cues After-hours motion in restricted zones Unauthorised back-store entries Door-open / cash-drawer correlation	04 Operations & Compliance Opening & closing procedure audit Mannequin / dummy-presence at promo zones Camera health & obstruction monitoring Daily / weekly executive summary email
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QSR EDITION

From Counter to Drive-Thru

Quick-service restaurants run on three numbers: speed, accuracy and food safety. Every camera in the dining room, kitchen and drive-thru already sees them. RetailTrack turns them into a live signal.

— QSR CAPABILITIES

Speed, Accuracy, Safety — Live.

Three operational levers every QSR brand tracks weekly via spreadsheets and mystery-shopper audits. EdgeVision tracks them per-shift, per-station, per-car — automatically.

1

Speed of Service

Drive-thru wait time per lane, counter queue length, mobile-pickup dwell, time-to-first-bag — every daypart benchmarked against itself, not a corporate average.

2

Kitchen Choke-Points

Idle stations, expo bottleneck, fry-station coverage, grill hand-off delays — flagged the moment they exceed threshold, not after the rush has cost a guest.

3

Food Safety & Hygiene

Gloves, hairnets, handwash frequency, holding-time at the heat lamp, spillage detection — scene-aware checks, low false-positive, ready for the next FSSAI / health audit.

— DRIVE-THRU OPERATIONS

Every Lane. Every Car. Every Second.

The drive-thru is the single biggest revenue line at most QSR sites — and the hardest to measure. EdgeVision instruments it end-to-end on the cameras you already have.

- Live per-lane queue length & wait time
- Order-board → handoff-window timing per car
- Pull-forward, blocked-lane & 'driver not served' alerts
- Lane-utilisation balance during peak hours
- Daypart vs forecast adherence



— **MORE OUT-OF-THE-BOX · QSR**

Every Camera, Every Service Window.

01 Counter & Kiosk Queue length vs labour deployed by daypart Self-order kiosk abandonment & completion Greeter / smile-at-window compliance Mobile-order pickup-bay dwell & handoff	02 Kitchen & Holding Station coverage per shift (grill · fry · expo) Hot-hold & cold-hold zone compliance Order assembly time & bagging accuracy Wipe-down cadence between rushes	03 Cleanliness & Safety Dining-area spillage & tray-bus cadence Restroom check cadence vs schedule Slip-and-fall hazard auto-flagging End-of-day closing-audit photo trail	04 Compliance & Audit PPE: gloves · hairnet · apron per station Handwash frequency vs corporate SOP Camera health & obstruction monitoring Daily / weekly executive summary email
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CHAPTER 05

The Paradigm Shift

Why GenAI-powered analytics finally makes the in-store camera a primary business sensor.

THE PARADIGM SHIFT

Why GenAI Analytics Changes Everything.

DIMENSION	TRADITIONAL CV	EDGEVISION GENAI
Detection	Bounding boxes & pixel thresholds	Contextual scene understanding
Accuracy	High false-positive in varying light	Adaptive to lighting & occlusions
Self-Service	Cannot determine purchase intent	Full customer journey with billing
Hygiene	Static detection, frequent false alerts	Scene-aware: items vs. decor
Adaptability	Manual re-calibration per store	Self-adapting, zero retraining
Insights	Counts and timestamps only	Rich descriptions: why, not what

“Traditional analytics tells you someone entered the store. EdgeVision tells you they browsed for 4 minutes, showed interest in the brand wall, and left without purchasing — and which staff member was nearest.”

— HYBRID INTELLIGENCE

Real-Time Vision + On-Device GenAI.

1

Hybrid Intelligence

Vision AI runs continuously at 15+ FPS. GenAI (VLM) is triggered only when the rule engine demands context — about 90% compute reduction.

2

Zero Cloud Dependency

Everything happens on Intel edge compute inside the site. No frames, no biometrics, no PII transmitted off-premise. Full data sovereignty.

3

Configurable Rules

Node-RED-style rule engine. Tune thresholds (e.g. '> 4 staff in back store') without code changes or retraining.

15₊

FPS CONTINUOUS VISION AI PER CAMERA

~90%

COMPUTE REDUCTION VIA VLM TRIGGERING

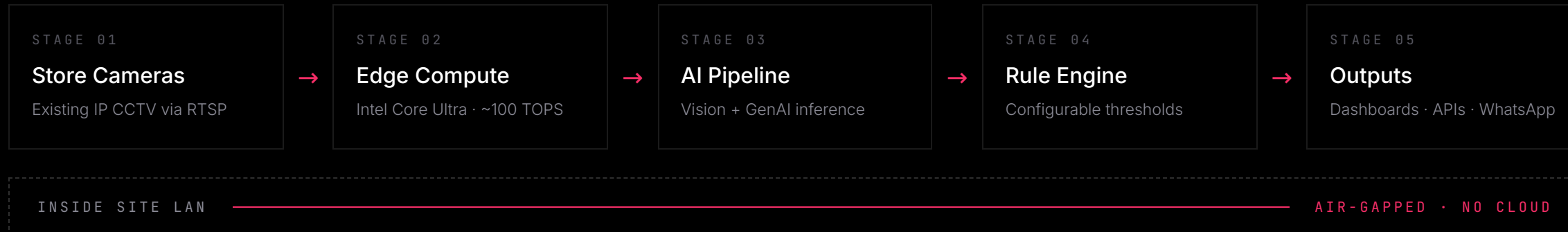
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FRAMES, BIOMETRICS OR PII OFF-PREMISE

EDGE-FIRST ARCHITECTURE

Built for the Store LAN.

High-performance Intel edge compute on the site LAN. Reads existing CCTV via RTSP. Runs the full pipeline locally.



HARDWARE & PRIVACY

Intel-Powered. Privacy by Design.

HARDWARE · REFERENCE EDGE NODE

PLATFORM	Lenovo · Model TBD
PROCESSOR	Intel Core Ultra 9 285H · ~100 TOPS
MEMORY	32 GB
STORAGE	1 TB NVMe
OS	Ubuntu 24.04 LTS
NETWORK	Closed LAN, isolated

PRIVACY & SECURITY

- ✓ On-prem deployment — all processing inside the site
- ✓ Air-gapped — no external API calls required
- ✓ Role-based access (RBAC) for managers & HQ
- ✓ FIFO video retention with configurable policy
- ✓ Tamper-resistant edge appliance
- ✓ Zero biometric data leaves the device

ENGAGEMENT MODEL

From CCTV to Insights in Three Weeks.

WEEK 01

Audit & Install

Site survey, camera audit, RTSP setup. Intel edge compute installed on the site LAN. Baseline data collection starts.

WEEK 02

Calibrate & Tune

Zone definitions, threshold tuning per fixture, alert routing setup, custom rules wired to messaging stack.

WEEK 03

Handover & Train

Manager training, dashboard onboarding, alert rehearsal, first weekly executive insights report delivered.

What you get at end of week 3: a fully running EdgeVision installation, two weeks of baseline analytics, a configured alert pipeline, and an operations team that has run a full daily cycle on the new console.

ABOUT

The Team Behind EdgeVision.

Pattern AI Labs builds intelligent edge AI for enterprise operations — democratising real-time video intelligence with privacy and actionable insights in seconds.

01

Generative AI & CV

Multi-modal models for scene understanding, contextual reasoning and visual intelligence — tuned for enterprise edge.

02

Edge AI Optimisation

Inference optimised for Intel NPU + iGPU on Core Ultra series. OpenVINO toolchain. Performance & thermal-aware.

03

Real-Time Anomaly

Instant alerts for unusual activity, safety violations and compliance — backed by configurable rule logic.

04

Enterprise Integration

REST APIs, webhooks, WhatsApp, dashboard plugins. Integrates with POS, BI stack & loyalty platform.

INTEL EDGE AI PARTNER SPOTLIGHT Featured as a leading edge-AI innovator in Intel's exclusive partner ecosystem. Optimised on Intel hardware for real-time inference at the edge.

Let's bring intelligence to every store & line.

A focused 3-week pilot. Same cameras. New decisions. Measurable lift in customer experience, staff productivity and QSR speed-of-service.

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